MATH 564

VALVE DESIGN

(21)	1.95	1.63	1.00	71.2
(22)	1.39	1.97	2.27	fail
(23)	5	1	4	0.8

	Armijo Backtracking	Strong Wolfe Alg 3.5 & 3.6
Gradient Descent	YES $0<c_1<1$ $0<\rho<1$	YES $0<c_1<c_2<1$
Conjugate Gradient	NO	YES * $0<c_1<c_2<1/2$
BFGS	NO	YES * $0<c_1<c_2<1$
SR1	NO	NO

typical parameter values

$\rho = 1/2$ $\quad c_1 = 0.0001$ $\quad c_2 = 0.9$

$c_2 = 0.45$

* restart conditions

Finding Roots of systems of Equations (Nonlinear)

$F : \mathbb{R}^n \to \mathbb{R}^m$

$$F_1(x) = \cdots$$
$$F_2(x) = \cdots$$
$$\vdots$$
$$F_m(x) = \cdots$$

Define $G : \mathbb{R}^n \to \mathbb{R}$ as $G(x) = F_1(x)^2 + F_2(x)^2 + \cdots + F_m(x)^2$

we have $\nabla G(x) = 2F_1(x)\nabla F_1(x) + \cdots + 2F_m(x)\nabla F_m(x)$

$$\nabla G(x) = 2\begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_2}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_1} \\ \frac{\partial F_1}{\partial x_2} & \frac{\partial F_2}{\partial x_2} & & \\ \vdots & & & \\ \frac{\partial F_1}{\partial x_n} & & & \end{pmatrix}\begin{pmatrix} F_1 \\ F_2 \\ \vdots \\ F_m \end{pmatrix}$$

$$\nabla G(x) = 2J^T F$$